

at least l periods relative to hospitalization by its sample analog. Third, we form IW estimates $\widehat{v}_\ell$ by taking weighted averages of $\widehat{\delta}_{e,\ell}$ (returned from step one) with sample cohort share (returned from step two) as weights.

In Table 3, we report the FE estimates $\widehat{\mu}_\ell$ and the IW estimates $\widehat{v}_\ell$, as well as the underlying $CATT_{e,\ell}$ estimates $\widehat{\delta}_{e,\ell}$. In this application, the FE estimates and IW estimates happen to be very similar in their magnitude. The conclusion from Dobkin et al. (2018) based on FE estimates similar to ours still holds: we find a substantial and persistent decline in the earnings due to hospitalization, and the increase in out-of-pocket spending is transitory and small in comparison.

However, the FE estimates still suffer from non-convex and non-zero weighting as we show in Section 5.3, which can lead to interpretability issues depending on the amount of treatment effect heterogeneity. Even though for both outcomes $\widehat{\mu}_0$ falls in the convex hull of its underlying $CATT_{e,0}$ estimates, $\widehat{\mu}_{-2}$ turns out to be outside the convex hull of its underlying $CATT_{e,-2}$ estimates. In contrast, by construction, the IW estimates $\widehat{v}_\ell$ fall within the convex hull of its underlying $CATT_{e,\ell}$ estimates and are unaffected by $CATT_{e,\ell'}$ estimates from other periods $\ell' \neq \ell$. Thus, they have an interpretation as an average effect of the treatment on the treated ℓ periods after initial treatment.

6 Conclusions

This paper analyzes the behavior of relative period coefficients μ_ℓ on the indicator for being ℓ periods away from the treatment from two-way fixed effects regressions in settings with variation in treatment timing and treatment effects heterogeneity. For dynamic treatment effects, researchers are usually interested in estimating some average of treatment effects from ℓ periods relative to the treatment, and it is common to report the coefficient estimate $\widehat{\mu}_\ell$ assuming that interpretation is valid. However, we show that in the presence of heterogenous treatment effects, the coefficient μ_ℓ does not necessarily capture the dynamic treatment effect as it can fall outside the convex hull of $CATT_{e,\ell}$ from its corresponding period ℓ and may pick up spurious terms consisting of treatment effects from periods other than ℓ . This negative result is based on the decomposition of the coefficients μ_ℓ as a linear combination of cohort-specific average treatment effects on the treated $CATT_{e,\ell}$. The weights in the linear combination can be non-convex, and non-zero on relative periods $\ell' \neq \ell$.

Given these negative results on two-way fixed effects regression estimators, we propose “interaction-

weighted” (IW) estimators for estimating dynamic treatment effects. The IW estimators are formed by first estimating $CATT_{e,\ell}$ with a regression saturated in cohort and relative period indicators, and then averaging estimates of $CATT_{e,\ell}$ across e at a given ℓ . These $CATT_{e,\ell}$ are identified under parallel trends and no anticipation assumptions. These estimators are easy to implement and robust to heterogeneous treatment effects across cohorts; the IW estimator associated with relative period ℓ is guaranteed to estimate a convex average of $CATT_{e,\ell}$ using weights that are sample share of each cohort e .

Finally, we illustrate the empirical relevance of our results by estimating the dynamic effect of hospitalization on the out-of-pocket medical spending and labor earnings using a setup similar to Dobkin et al. (2018). We find non-convex and non-zero weights for two-way fixed effects regression in this example, and show that the resulting estimates indeed sometimes fall outside the convex hull of the underlying $CATT$ estimates due to contamination by treatment effects from other relative periods. IW estimates, on the other hand, are weighted averages of the underlying $CATT$ estimates with weights representative of cohort share.

More broadly, our paper suggests that researchers can do more in this context to assess the validity of underlying assumptions and how violations may impact their estimates. We demonstrate the sensitivity of two-way fixed effects regressions to underlying assumptions and provide researchers with tools to assess and address these issues. Specifically, for average treatment effects estimated using two-way fixed effects regressions we recommend that empirical researchers directly estimate the underlying weights on cohort-specific average treatment effects using our proposed auxiliary regressions. This exercise allows empirical researchers to assess the degree of potential contamination under any assumed structure of treatment effects heterogeneity. We also recommend empirical researchers consider estimation methods more robust to treatment effects heterogeneity such as the IW estimator we propose. Developing additional tools for applied researchers to identify and estimate interpretable parameters of interest is a promising avenue for future research.

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